

Shock: Pathophysiology and Management

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Learning Objectives

1. Understand the underlying pathophysiology of shock
2. Identify the unique differences in the presentation and pathophysiology of the four major categories of shock: hypovolemic, cardiogenic, obstructive, and distributive
3. Describe the clinical approach to a patient with shock and identify the next best steps in management
4. Understand and identify the pharmacologic treatments for each of the major categories of shock

Lecture Layout

- Measurements to Describe Shock
 - These concepts are discussed in the following lectures.
 - Hemodynamics and Circulation
 - Arterial Blood Pressure Regulation
 - Molecular Physiology of Vascular Endothelial Cells
 - Molecular Physiology of Vascular Smooth Muscle Cells
- Pathophysiology & Clinical Presentation of Shock
 - Hypovolemic
 - Cardiogenic
 - Obstructive
 - Distributive
- Management

Mechanisms to Describe Shock

- There are **five** main measurements used to determine what type of shock a patient is in:
 - 1. Post Capillary Wedge Pressure**
 - Measures the pressure in the **left atrium**
 - Increases with cardiac failure
 - 2. Cardiac Output**
 - Cardiac Output = stroke volume x stroke rate
 - 3. Systemic Vascular Resistance**
 - Measures the constriction of arterioles in the systemic vasculature
 - 4. Saturation of Venous O₂ (SVO₂)**
 - Measurement of amount of oxygen in the venous circulation
 - Assess the ability of tissues to **extract** oxygen from the arterioles
 - 5. Central Venous Pressure**
 - Assesses the amount of blood returning to the heart
 - Measures the amount of blood in the Right Atrium

- Lectures to Reference:
 - Lecture 17: Hemodynamics and Circulation
 - Lecture 28-29: Physiology of Cardiac Cycle Parts 1, 2

Post Capillary Wedge Pressure

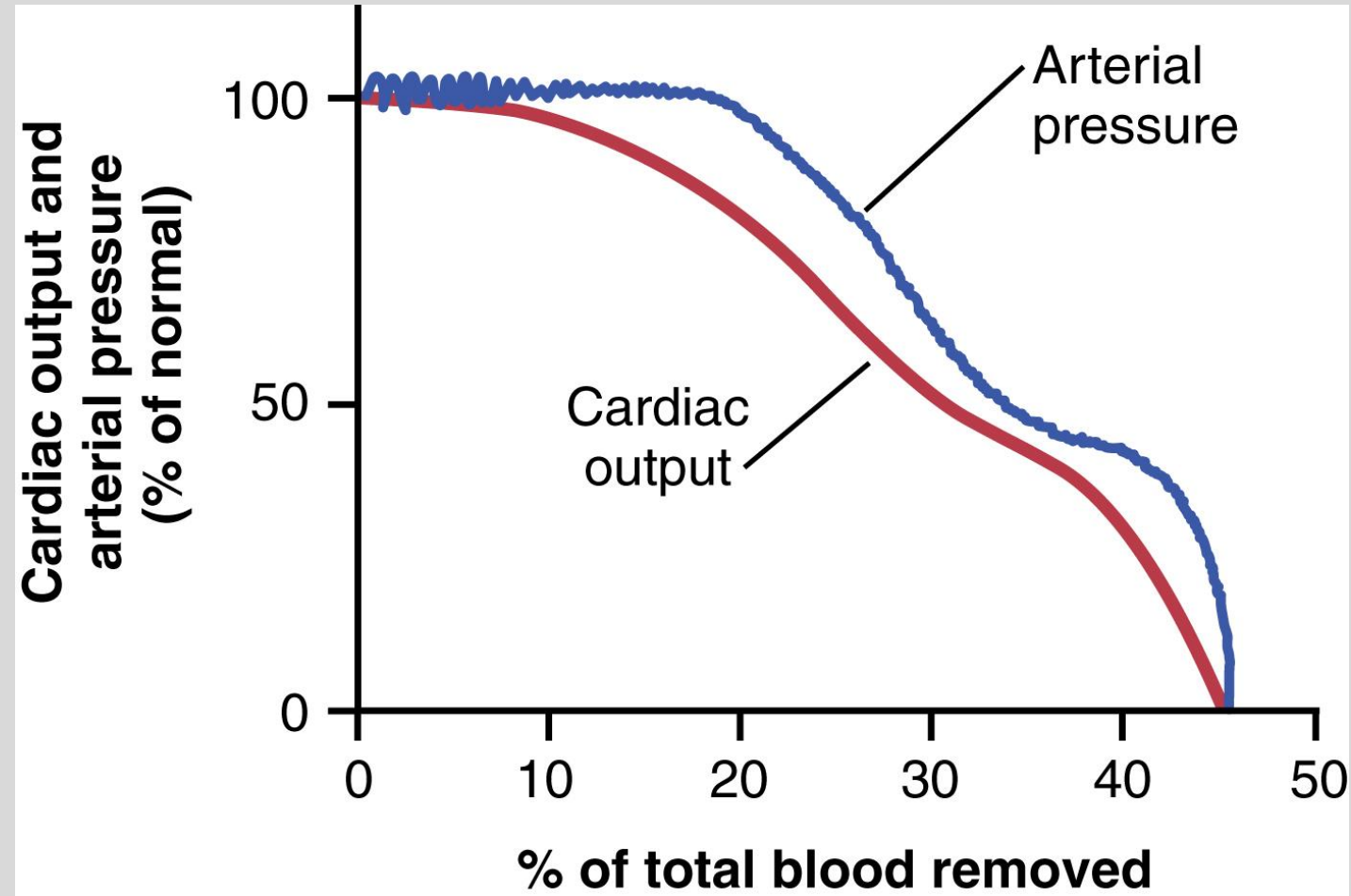
- Measurement of the pressure in the **Left Atrium**
- Key: helps distinguish the type of obstructive shock:
 - If there is an obstruction before the Left Atrium (like a PE or tension pneumothorax), the PCWP will be **LOW**
 - If there is an obstruction after the Left Atrium, or an obstruction that affects the entire heart (cardiac tamponade), the PCWP will be **HIGH**
- Measured with a Swann-Ganz catheter

Cardiac Output

- Formula is derived from the stroke volume and stroke rate
- All changes in cardiac output must be produced by changes in the heart rate and/or the stroke volume

$$CO = HR \times SV$$

- Cardiac output can be altered by sympathetic or parasympathetic responses
- Cardiac output can be altered by both preload and afterload of the LV



Total Peripheral Resistance

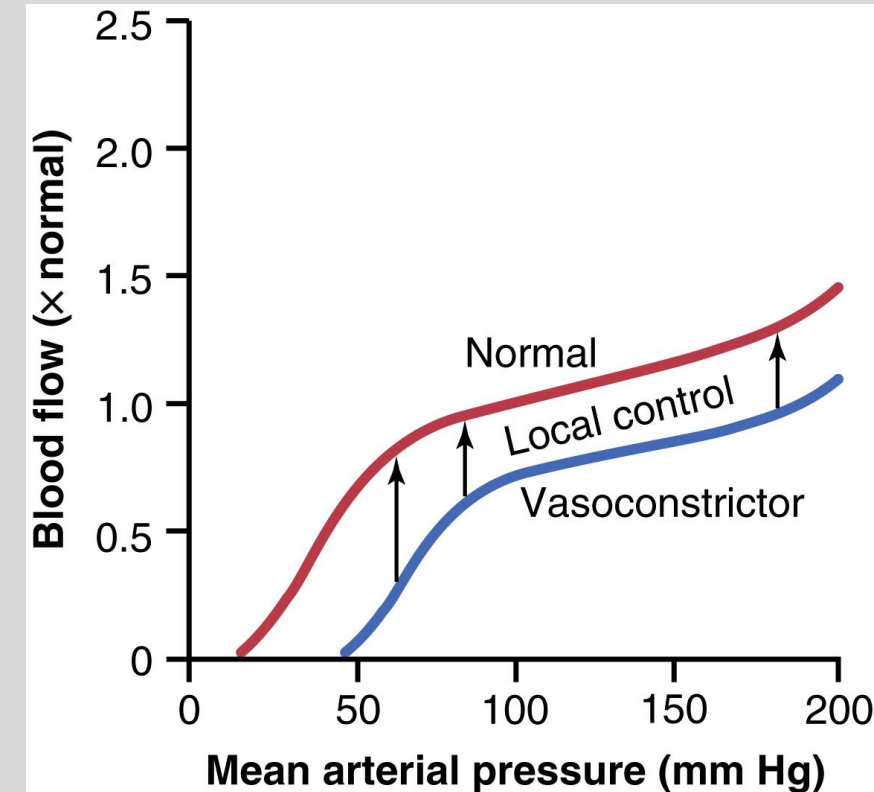
Systemic Vascular Resistance

- Measures the overall level of constriction in the arteriole system
- TPR/SVR is dictated by the resistance of the blood moving through arteries (Poiseuille's Law):

$$R = \frac{8Ln}{\pi r^4}$$

r = radius; L = length of vessel; n = viscosity of blood

- **Radius has the largest effect on TPR**
- **This is an important concept when we consider the effects of vasoconstriction!**

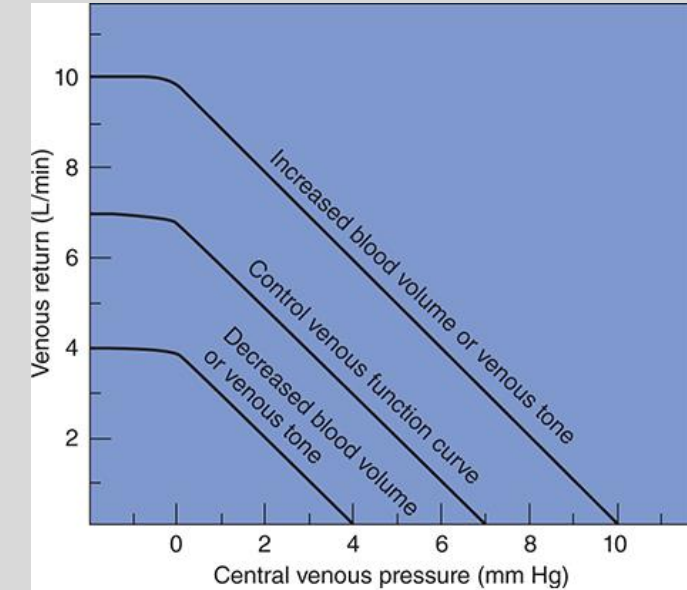


Saturation of Venous O₂

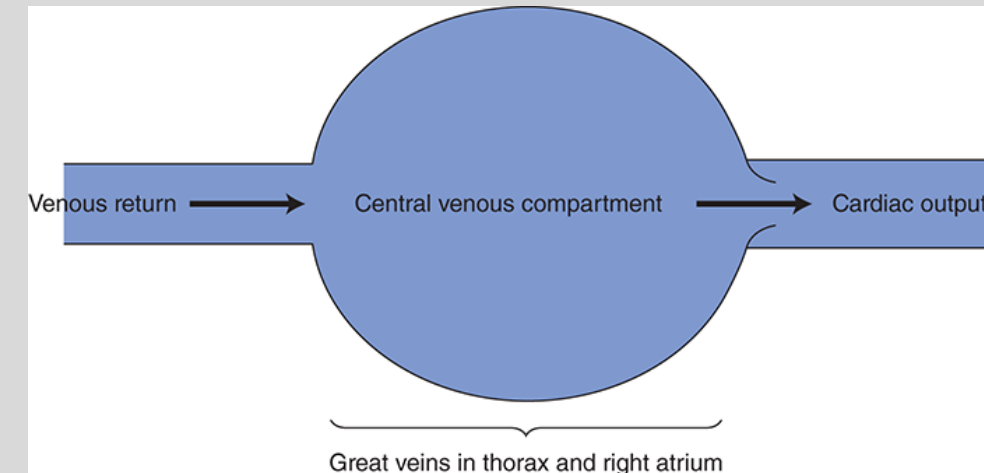
- Measurement of oxygen returning to the right heart
- Used to assess oxygen delivery and oxygen consumption and reflect tissue oxygenation
- Increased SVO₂ represents a decreased ability of tissues to extract oxygen from the blood
- Decreased SVO₂ represents an increased ability of tissues to extract oxygen from the blood

Central Venous Pressure

- Central venous pressure is a measurement for the right atrium
- CVP is a balance between two opposing forces:
 - 1) The ability of the heart to pump blood out of the RA and ventricle into the lungs
 - 2) The blood flow moving from the peripheral veins to the right atrium
 - Caused by increased blood volume
 - Increased large vessel tone
 - **Dilation of the arterioles**



Source: David M. Harris: *Mohrman and Heller's Cardiovascular Physiology, 10th Edition*
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Definition of Shock

- In general, shock occurs whenever there is a generalized, severe reduction in blood supply to the body tissues, and the needs of the tissues are not met
 - Even with cardiovascular compensatory mechanisms, arterial pressure is still low
 - Shock is typically due to one of two things: disturbances in cardiac output or systemic vascular resistance
- Adequate oxygen therapy is reliant on the following physiologic parameters: **cardiac output and systemic vascular resistance**
- In terms of blood pressure, think shock if systolic < 90 mmHg or MAP < 65 mmHg

Main Mechanisms of Shock

Types of Shock	Mechanism	Examples of Causes	Primary Insult
Cardiogenic	Impaired cardiac function	LV Systolic failure Valvular disorders Arrhythmias Right Ventricular Failure	↓ CO
Distributive – Vasodilatory	Impaired vasoconstriction	Sepsis Anaphylaxis Neurogenic Pancreatitis	↓ SVR
Hypovolemic	Volume depletion	Hypovolemia Hemorrhage	↓ Volume (↓ CO due to ↓ PCWP)
Obstructive	Impairment of cardiopulmonary blood flow	Pulmonary Embolism Cardiac Tamponade Tension Pneumothorax	↓ CO

**** The Highlighted Measurement is the primary insult**

- (2022) Approach to the Patient with Shock. Harrison's Principles of Internal Medicine.
- Adapted from Shock and Hemodynamics in the CCU with Dr. Brown, Louisville Medical Lecture Series.

Hypovolemic Shock

- Cause: Loss of blood or plasma volume (~20% blood loss)
 - Hemorrhage
 - Severe vomiting or diarrhea
 - Heat stroke leading to fluid depletion
 - Severe burns
 - Third spacing: shift of fluid from the intravascular space to the interstitial space (pancreatitis, cirrhosis, postoperatively)
- Patient Presentation:
 - Narrow pulse pressure
 - Cool and clammy extremities
 - Poor capillary refill
 - Flat jugular neck veins
 - Dry mucous membranes

• (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.

• (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology

• (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Hypovolemic Shock: Pathophysiology

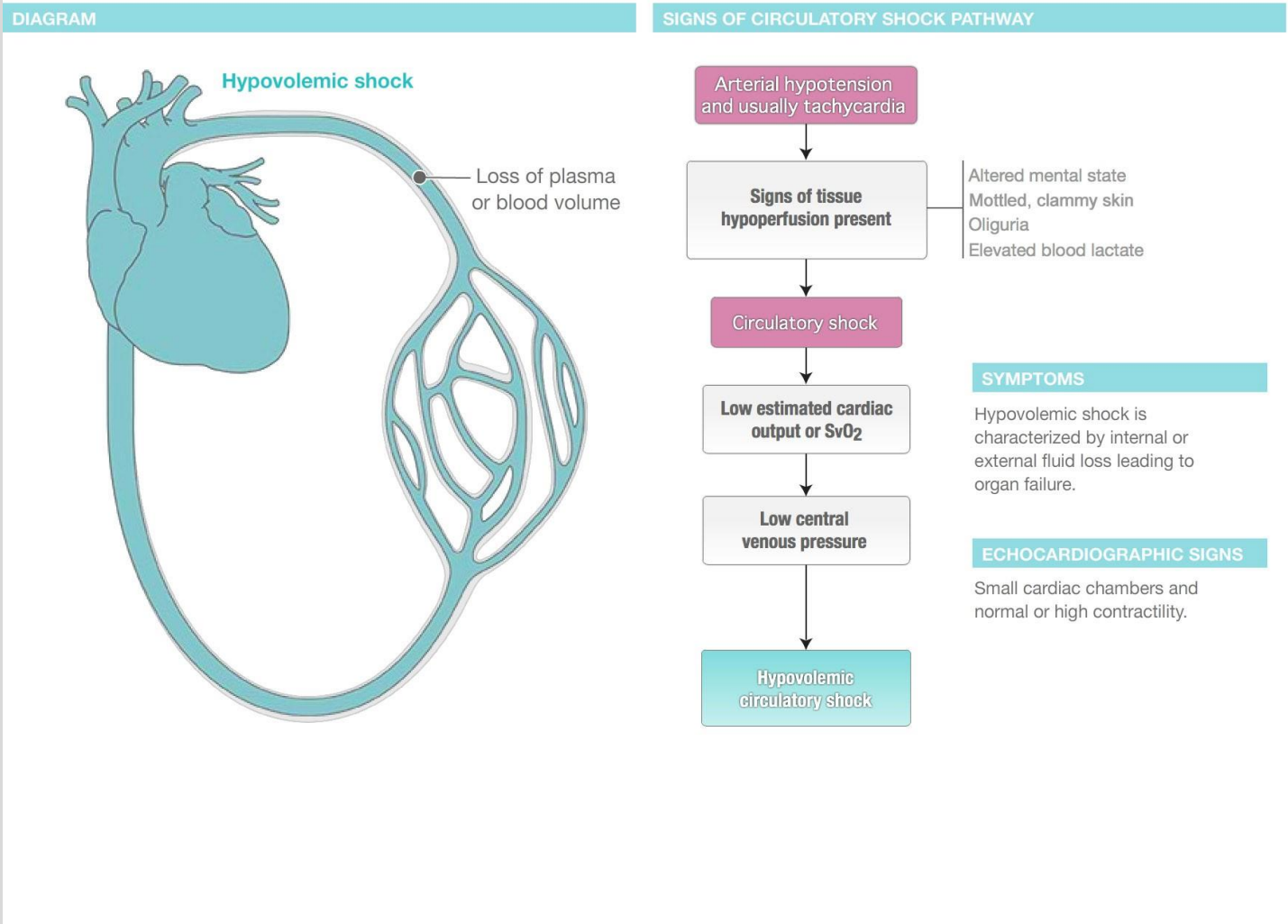
- Hypovolemia leads to **REDUCED CARDIAC FILLING** and **REDUCED STROKE VOLUME**
 - **Preload is decreased**
- **Central Venous Pressure:** decreased
 - Reduced overall volume will decrease amount of blood in RA
- **PCWP: decreased**
 - Reduced overall volume will decrease amount of blood in LA
 - Decreased preload
- **Cardiac Output:** decreased (volume depletion)
 - Reduced stroke volume will reduce overall CO
- **Systemic Vascular Resistance:** increased
 - Baroreceptors will sense the decreased mean arterial pressures, increasing vasoconstriction
- **SVO₂:** decreased
 - Higher SVR will slow the flow of blood through the capillaries, increasing the amount of time for gas exchange

• (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.

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Hypovolemic Shock: Diagram



Cardiogenic Shock

- Cause: Decreased myocardial contractility
 - Acute Myocardial infarction
 - Viral or drug induced cardiomyopathy
 - Decompensated CHF (congested heart failure)
 - Acute or end stage valvular disease
 - Dysrhythmia (due to decreased filling time)
- Patient Presentation:
 - Narrow pulse pressure
 - Cool and clammy extremities
 - Poor capillary refill
 - Elevated jugular venous pressure
 - Positive hepatojugular reflex
 - S3 gallop
 - Pulmonary rales

• (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.

• (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology

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Cardiogenic Shock: Pathophysiology

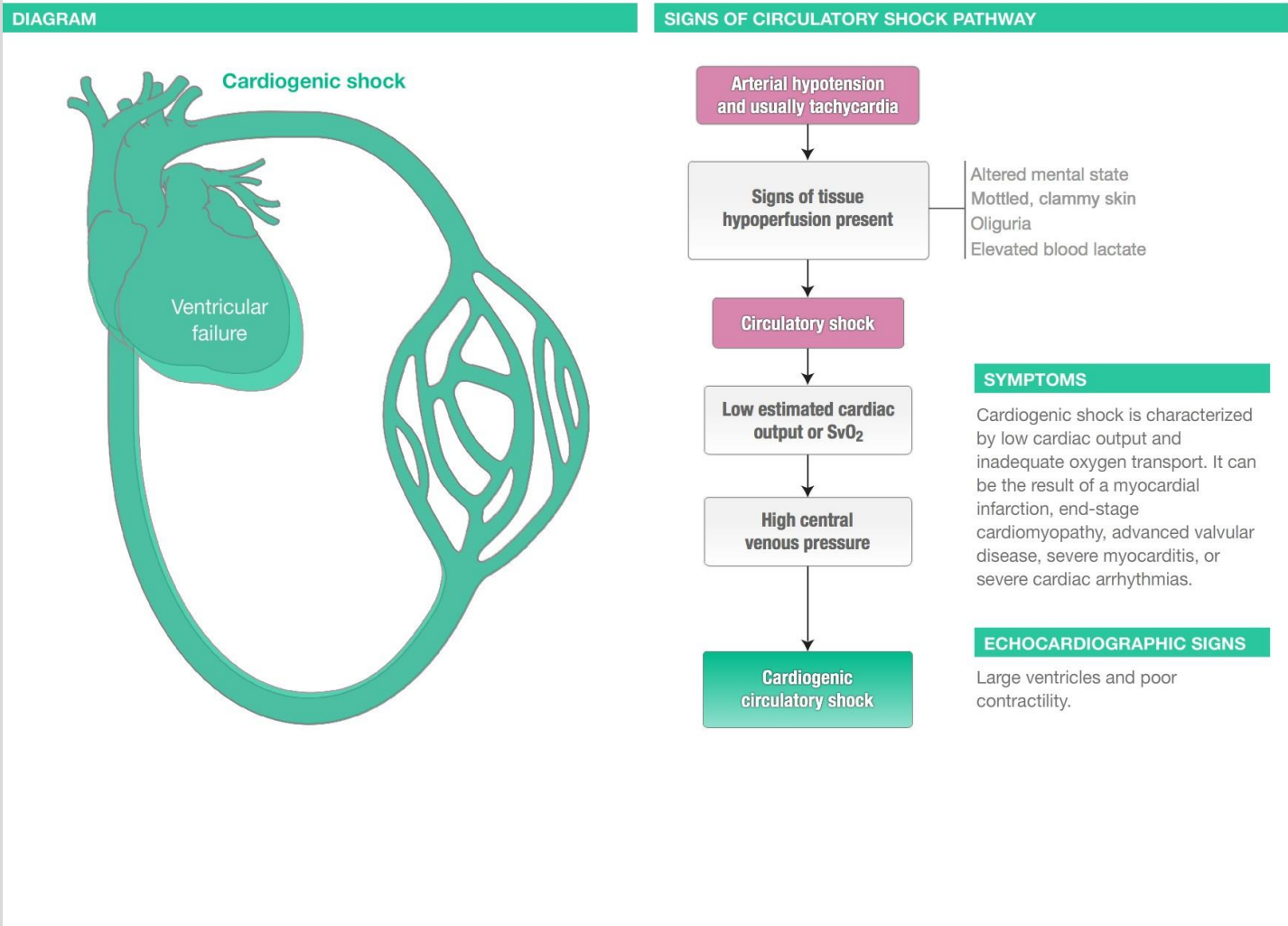
- Cardiac failure leads to **INCREASED PRELOAD** and **REDUCED CARDIAC OUTPUT**
- **Central Venous Pressure:** increased
 - Will have fluid accumulation behind the heart (**Right Heart/Venous System**)
- **PCWP:** increased
 - Preload is increased, but stroke volume decreased
 - Will have fluid accumulation behind the heart (**LEFT ATRIUM**)
- **Cardiac Output:** decreased
 - Reduced stroke volume will reduce overall CO (decreased ejection fraction)
- **Systemic Vascular Resistance:** increased
 - Baroreceptors will sense the decreased mean arterial pressures, CNS will promote vasoconstriction
- **SVO₂:** decreased
 - Higher SVR will slow the flow of blood through the capillaries, increasing the amount of time for gas exchange

• (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.

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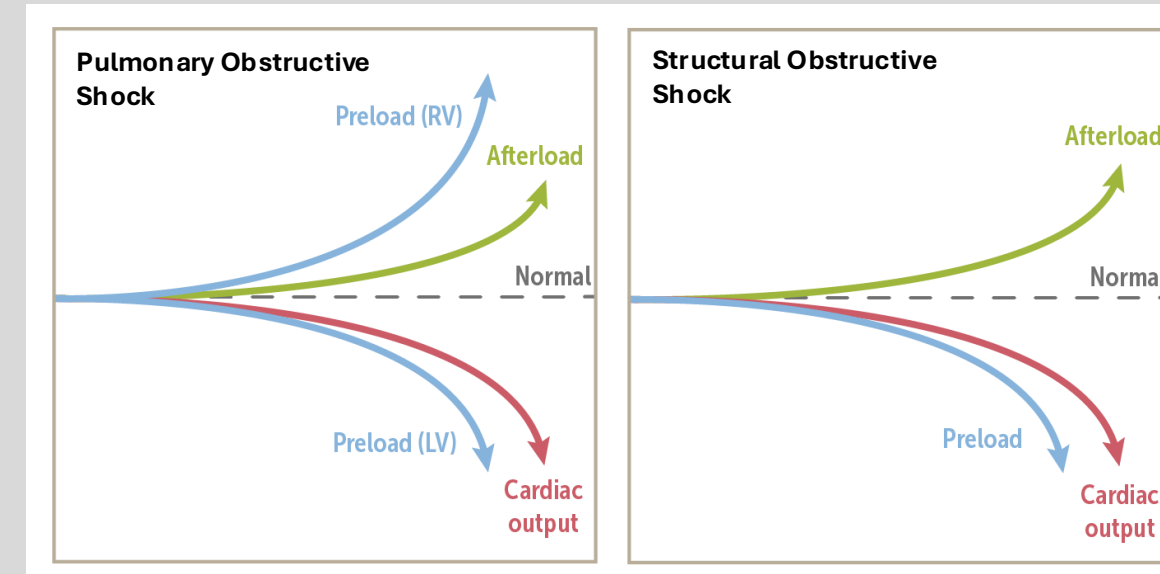
• (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Cardiogenic Shock: Diagram



Obstructive Shock

- Causes:
 - **Pulmonary Obstructive Shock ($\uparrow$ RV afterload & $\downarrow$ LV Preload)**
 - Increased resistance within the pulmonary circulation prevents blood from reaching the left side of the heart
 - Pulmonary Embolism
 - Tension Pneumothorax
 - **Structural Obstructive Shock ($\downarrow$ RV Preload)**
 - Restricted diastole/filling
 - Cardiac tamponade
 - Constrictive pericarditis



ScholarRx

- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease
- (2025) Management of Shock. ScholarRx

Obstructive Shock

- Patient Presentation:
 - Narrow pulse pressure
 - Cool and clammy extremities
 - Poor capillary refill
 - **Tension Pneumothorax:** absent breath sounds
 - **Cardiac Tamponade:** distant heart sounds, jugular venous distension (JVD), hypotension
 - **Pulmonary Embolism:** dyspnea and shortness of breath

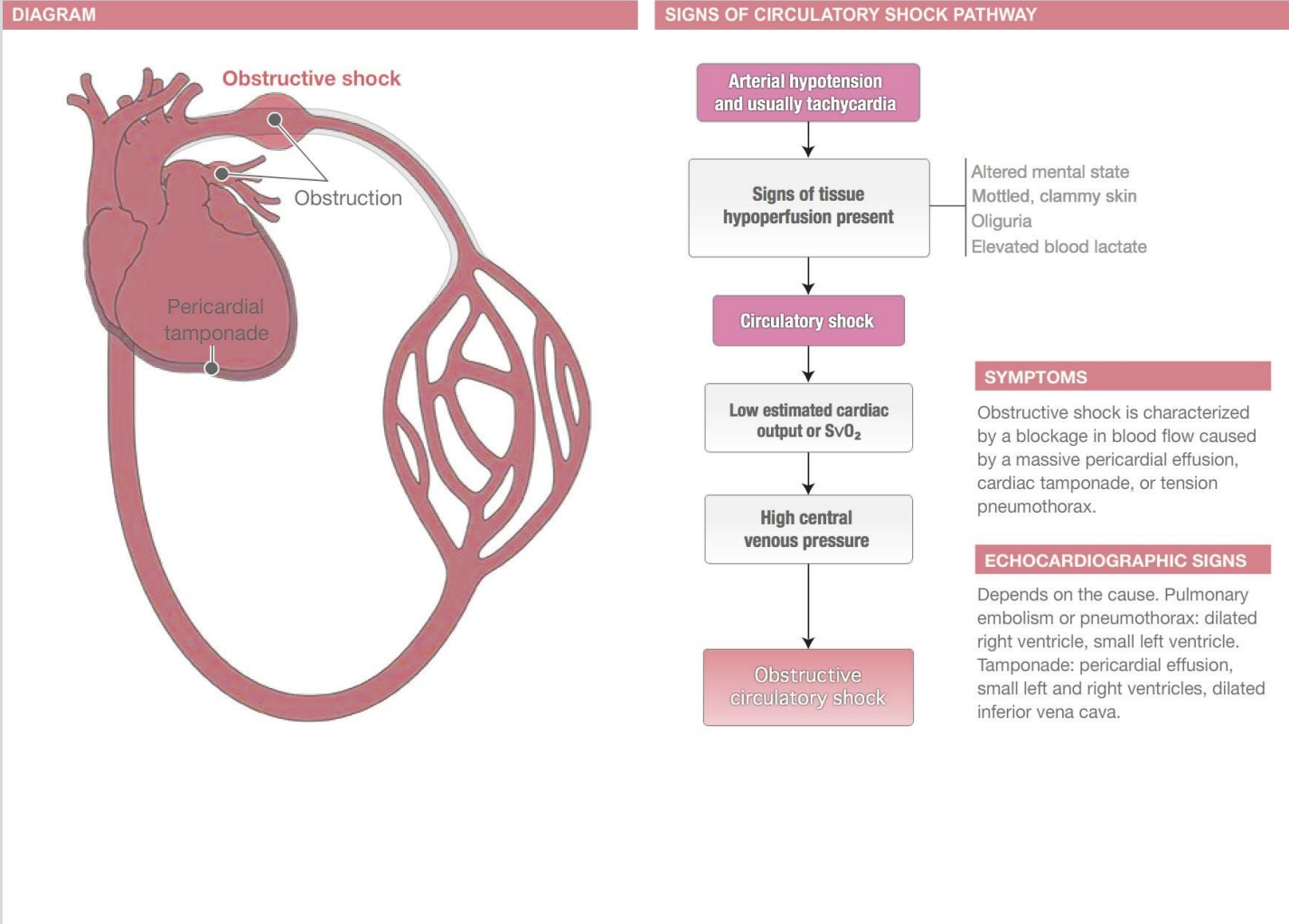
- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Obstructive Shock: Pathophysiology

- Obstruction leads to **DECREASED PRELOAD** and **REDUCED CARDIAC OUTPUT**
- **Central Venous Pressure:** increased
 - Will have fluid accumulation behind the heart
- **PCWP: VARIABLE**
 - Pulmonary Obstruction: **decreased**
 - Causes: pulmonary embolism
 - Blood is obstructed **BEFORE** entering the LA
 - Decrease LV Preload and Increase RV afterload
 - Cardiac Constriction: **increased**
 - Causes: Cardiac tamponade
 - The entire heart is constricted, reducing diastolic filling ability and decreasing stroke volume
 - Decrease RV preload and decrease diastolic filling
- **Cardiac Output:** decreased
 - Reduced stroke volume will reduce overall CO (decreased ejection fraction)
- **Systemic Vascular Resistance:** increased
 - Baroreceptors will sense the decreased mean arterial pressures, CNS will promote vasoconstriction
- **SVO₂:** decreased
 - Higher SVR will slow the flow of blood through the capillaries, increasing the amount of time for gas exchange

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- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease
- (2022) Approach to the Patient with Shock. Harrison's Principles of Internal Medicine.

Obstructive Shock: Diagram



Distributive Shock

- Causes: loss of vascular tone
 - **Sepsis**
 - **Anaphylaxis**
 - **Neurogenic***
- Patient Presentation:
 - Wide pulse pressure
 - Warm and dry skin
 - Loss of sympathetic tone (neurogenic)
 - Anaphylaxis Signs & Symptoms: hives, urticaria, stridor, airway compromise

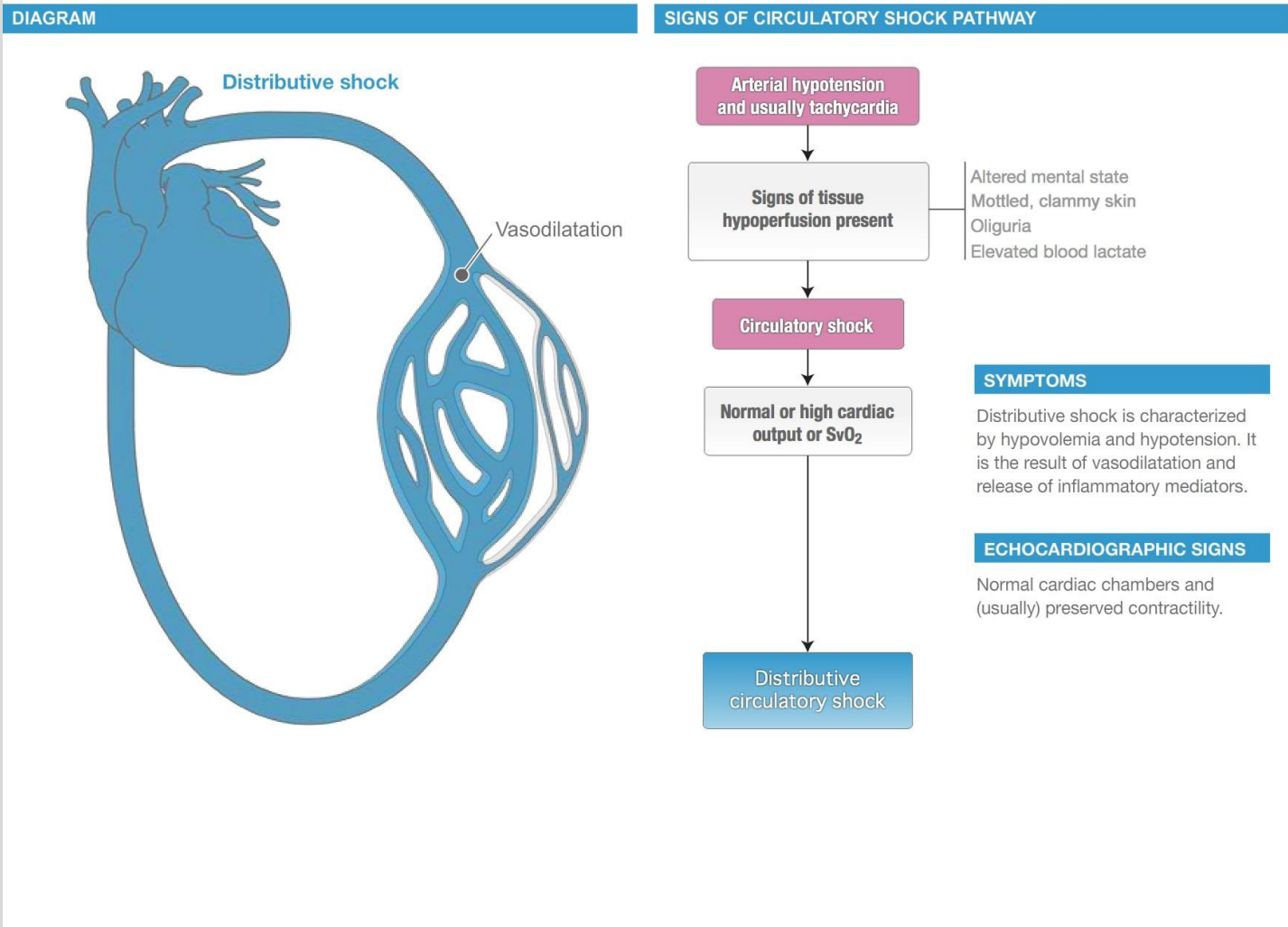
- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Distributive Shock: Pathophysiology

- Distributive shock is due to **loss of vascular tone**, leading to **DECREASED RESISTANCE**
- **Central Venous Pressure:** decreased
 - Vascular tone will also decrease in the venous system
- **PCWP:** Decreased
 - Increased vasodilation allows more blood to remain in the circulatory
- **Cardiac Output:** variable
 - **Septic & Anaphylaxis:** increased
 - Sympathetic response to the drop of vascular tone
 - **Neurogenic:** decreased
 - No sympathetic response due to injury to the CNS
- **Systemic Vascular Resistance:** decreased
 - Driving physiology change in distributive shock
- **SVO₂:** increased
 - Lower SVR due to vasodilation will increase the speed of blood traveling through capillaries
 - Increased rate of flow will decrease the amount of time gas exchange can occur

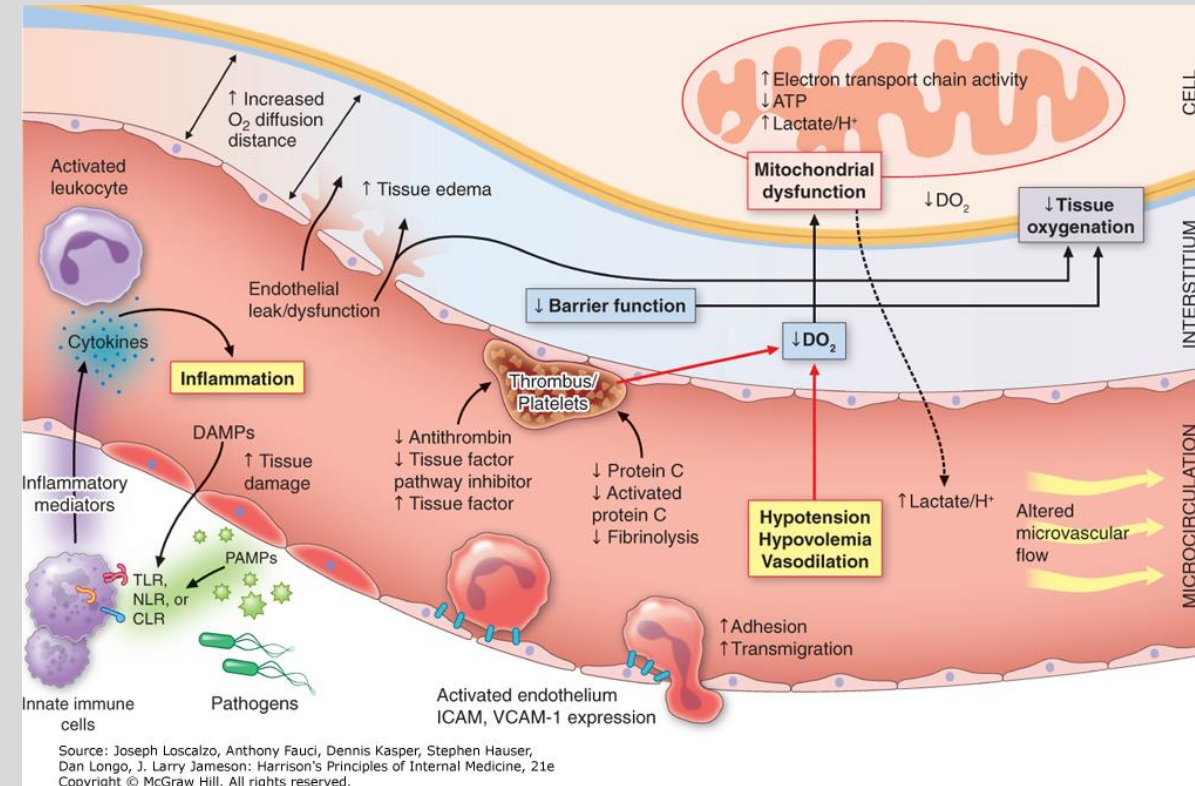
- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
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Distributive Shock: Diagram



Septic Shock: Pathophysiology

- Septic shock is usually caused by **gram-negative bacteria**, mediated by endotoxins known as lipopolysaccharide (LPS)
- LPS**, specifically the Lipid A portion binds to a pattern recognition receptor (PPR) called Toll Like Receptor 4 (TLR-4) on macrophages and B-Cells
 - Activating **TLR-4 stimulates the production of IL-1, TNF, and IL-6**
 - These cytokines promote **fever, weaken the bond between endothelial cells, and recruit further inflammation in the cells**
 - Nitric Oxide is also released, causing vasodilation**
 - LPS can also activate the coagulation cascade, causing disseminated intravascular coagulation via tissue factor production



- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
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Sepsis Criteria: SIRS

- SIRS criteria often used in hospital settings to assess systemic inflammation and/or sepsis
- **SIRS** is defined as meeting two of the four criteria:
 - Temperature over 38 or under 36 C
 - HR > 90
 - Respiratory Rate > 20 breaths per minute or partial pressure of CO₂ less than 32 mmHg
 - Leukocyte count > 12000 or less than 4000 OR over 10% immature bands
- **Sepsis:** Life-threatening organ dysfunction caused by dysregulated host response to infection
- **Severe Sepsis:** sepsis with one or more end-organ failures
- **Septic Shock:** hemodynamic instability despite intravascular volume repletion
 - **Defined as** sepsis and vasopressor therapy needed to elevated **MAP > 65 and lactate > 2 mmol/L**

Box 1. SIRS (Systemic Inflammatory Response Syndrome)

Two or more of:

Temperature >38°C or <36°C

Heart rate >90/min

Respiratory rate >20/min or PaCO₂ <32 mm Hg (4.3 kPa)

White blood cell count >12 000/mm³ or <4000/mm³
or >10% immature bands

From Bone et al.⁹

Early Sepsis: qSOFA & NEWS

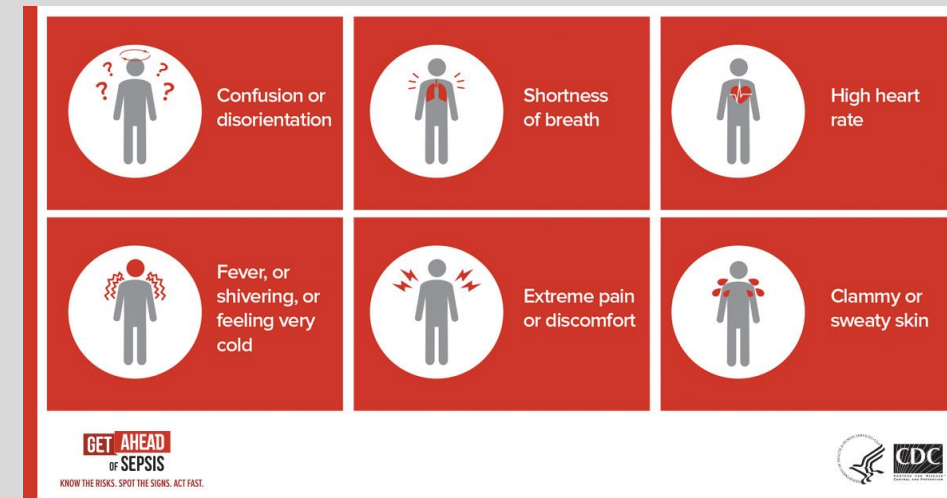
- **Sequential (Sepsis-Related) Organ Failure Assessment (SOFA)**
- **SOFA** is defined as meeting two of the three criteria:
 - Respiratory rate > 22 per minute
 - Altered mentation
 - Systolic BP < 100
- SOFA helps warn clinicians to further investigate organ dysfunction
- **National Early Warning Score**
- Based on a points system of the following physiologic parameters
 - Respiratory rate
 - Oxygen saturation
 - Systolic BP
 - Pulse
 - Level of consciousness/confusion
 - Temperature
- 0-4: low risk
- 5-6: medium risk
- 7+ : high risk

Box 4. qSOFA (Quick SOFA) Criteria

Respiratory rate $\geq 22/\text{min}$

Altered mentation

Systolic blood pressure ≤ 100 mm Hg



Sepsis Treatment:

- Broad spectrum antibiotics while awaiting results of blood cultures:
 - Goal: initiate IV antibiotic therapy within the first hour after diagnosis of sepsis or septic shock
 - Abx include: vancomycin + cefepime or piperacillin
- Hypotension:
 - Begin with IV fluid repletion, particularly crystalloid solutions
 - If fluids are ineffective, can treat with vasopressors
- **Key:** identifying the source of infection and removing it (if foreign body) will increase success rates

Surviving Sepsis
Campaign

BUNDLE

HOURL-1 BUNDLE: INITIAL RESUSCITATION FOR SEPSIS AND SEPTIC SHOCK:

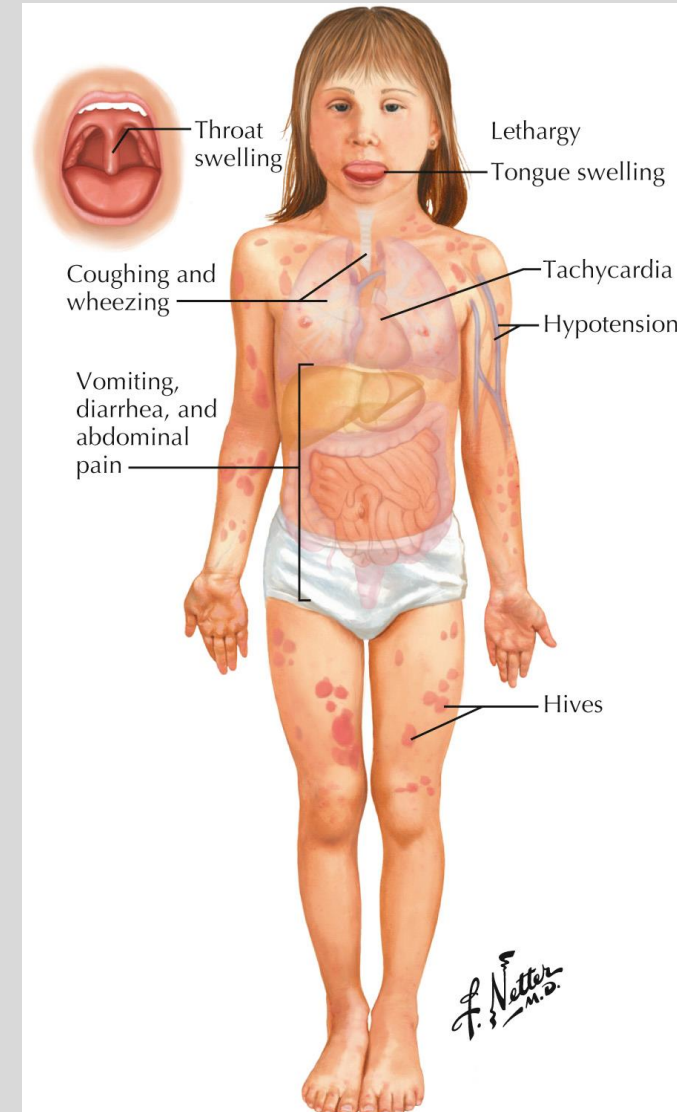
- 1) Measure lactate level.*
- 2) Obtain blood cultures before administering antibiotics.
- 3) Administer broad-spectrum antibiotics.
- 4) Begin rapid administration of 30mL/kg crystalloid for hypotension or lactate ≥ 4 mmol/L.
- 5) Apply vasopressors if hypotensive during or after fluid resuscitation to maintain a mean arterial pressure ≥ 65 mm Hg.

*Remeasure lactate if initial lactate elevated (> 2 mmol/L).

Source: Gregory A. Schmidt, John P. Kress, Ivor S. Douglas:
Hall, Schmidt and Wood's Principles of Critical Care, 5th Edition
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Anaphylactic Shock: Pathophysiology

- Anaphylaxis is an allergic condition where arterial pressure decrease drastically
- Caused by an antigen-antibody reaction
- This reaction causes a cascade of effects, resulting in mast cells in the periphery and basophils in the blood to release **histamine or histamine like-substances**
 - Increase vascular capacity and venous dilation
 - Dilation of the arterioles
 - Increased capillary permeability, with loss of fluid and protein into tissue spaces
- From a physiologic presentation, it is identical to septic shock



- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Anaphylactic Shock: Treatment

- The most important treatment in anaphylaxis is **epinephrine IM**
- Additional treatment should include:
 - Normal Saline to treat hypotension
 - Oxygen
 - Albuterol: help decrease bronchospasm, especially if present after epinephrine

Beneficial effects and adverse effects of epinephrine in the treatment of anaphylaxis

Beneficial effects	
At alpha-1 receptor:	
Increased vasoconstriction (at low doses)	
Increased peripheral vascular resistance	
Increased blood pressure	
Decreased mucosal edema (eg, in larynx)	
At beta-1 receptor:	
Increased heart rate (chronotropy)	
Increased force of cardiac contraction (inotropy)	
At beta-2 receptor:	
Decreased mediator release from mast cells and basophils	
Increased bronchodilation	
Increased vasodilation	
Adverse effects*†Δ	
Common and transient	Anxiety, palpitations, pallor, tremor, fear, restlessness, dizziness, headache
Uncommon (typically occur after overdose)	Ventricular arrhythmias, angina, myocardial infarction, pulmonary edema, sudden sharp increase in blood pressure, intracranial hemorrhage

* Risk of adverse effects may be increased in the following conditions:

- Use of tricyclic antidepressants, monoamine oxidase inhibitors, or cocaine.
- Some preexisting cardiovascular, central nervous system, or thyroid diseases. Examples include intracranial surgery, acute aneurysm, or untreated hyperthyroidism.

† Serious adverse effects such as those listed in the table potentially occur when epinephrine is given in overdose by any route, most commonly after an intravenous bolus injection, an overly rapid intravenous infusion, or an erroneous intravenous injection of a 1 mg/mL (1:1000) epinephrine solution instead of an appropriately diluted 0.1 mg/mL (1:10,000) or a 0.01 mg/mL epinephrine solution.

Δ Anaphylaxis can present as an acute coronary syndrome, arrhythmias, myocardial infarction, or angina, before or in the absence of epinephrine injection. This potentially occurs in patients with known coronary artery disease, patients in whom subclinical coronary artery disease is unmasked, and patients (including children) with transient coronary artery vasospasm in whom no cardiovascular abnormalities can be detected by electrocardiogram or echocardiography after recovery from anaphylaxis.

Neurogenic Shock

Neurogenic shock is due to dysregulation of the autonomic nervous system, causing a lack of sympathetic tone and unopposed parasympathetic response

Causes include:

- Deep general anesthesia: depressing the vasomotor center enough to cause vasomotor paralysis
- Spinal anesthesia: blocks sympathetic nervous outflow (especially damage above the T6 level)
- Brain Damage: damage to the basal regions/vasomotor neurons
- Will present similarly to septic/anaphylactic shock, except there will be no tachycardia

- (2020) Shock. Netter's Integrated Review of Medicine: Pathogenesis to Treatment.
- (2021) Circulatory Shock and Its Treatment. Guyton and Hall Textbook of Medical Physiology
- (2023) Hemodynamic disorders, thromboembolic disease, and shock. Robbins and Cotran Pathologic Basis of Disease

Results of Shock

Types of Shock	Examples of Causes	CVP	PCWP	CO	SVR	SVO ₂
Hypovolemic	- Hypovolemia - Hemorrhage	↓	↓	↓	↑	↓
Cardiogenic	- LV Systolic failure - Valvular disorders - Arrhythmias - RV Failure	↑	↑	↓	↑	↓
Obstructive - Decreased LV preload - Increased RV afterload	Pulmonary Embolism	↑	↓	↓	↑	↓
Obstructive Decreased RV preload	Cardiac Tamponade	↑	↑	↓	↑	↓
Distributive Sympathetic drive intact	Sepsis Anaphylaxis - Liver failure - Pancreatitis - Systemic inflammation	↓	↓	↑	↓	↑
Distributive No sympathetic drive	Neurogenic	↓	↓	↓	↓	↑

• (2023) Cardiovascular Function in Pathological Situations. Harris D.M.(Ed.), *Mohrman and Heller's Cardiovascular Physiology, 10th Edition*.
 • (2022) Sepsis and Septic Shock. Harrison's Principles of Internal Medicine.

Shock: Clinical Presentation

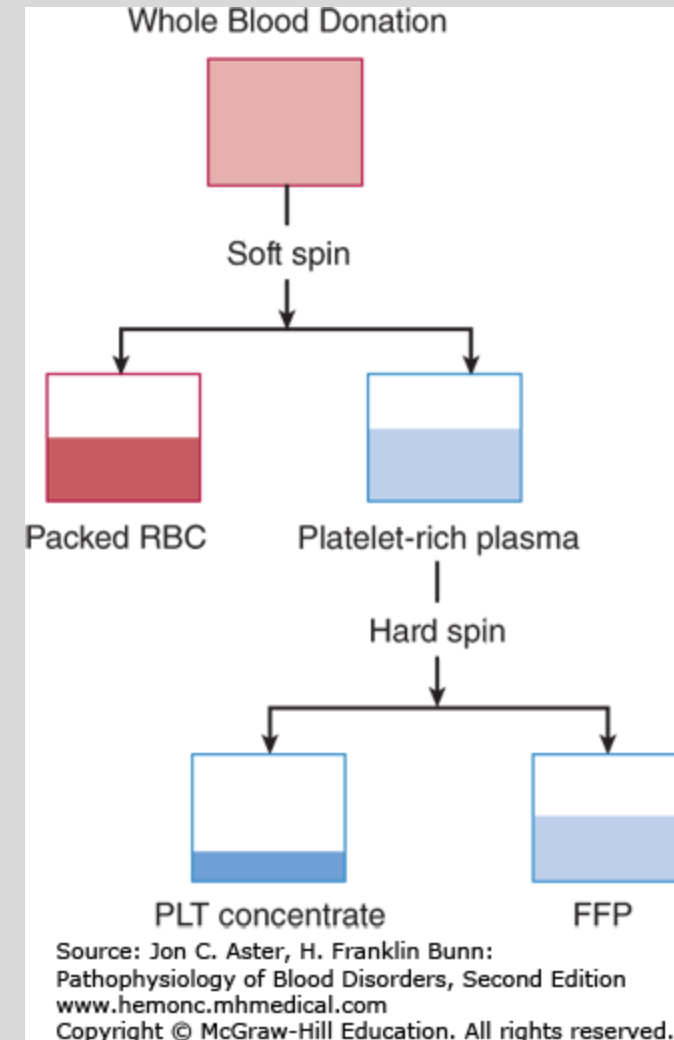
- Hypotension
- Tachycardia
- Bradycardia
- Tachypnea
- Altered mentation, agitation
- Hypoxemia
- Oliguria or anuria
- Cool extremities (**hypovolemic, cardiogenic, or obstructive shock**)
- Warm extremities (**septic**)
- Acrocyanosis, blue/pale/discolored skin, slow capillary refill
- Peripheral edema (in cardiogenic/obstructive shock)

Management of Shock

Types of Shock	Mechanism	Examples of Causes	Primary Insult	Management
Cardiogenic	Impaired cardiac function	LV Systolic failure Valvular disorders Arrhythmias Right Ventricular Failure	↓ CO	• Inotropes • Further cardiac intervention
Distributive – Vasodilatory	Impaired vasoconstriction	Sepsis Anaphylaxis Neurogenic Pancreatitis	↓ SVR	• Fluid resuscitation • Antibiotics (septic), • IM epinephrine (anaphylaxis)
Hypovolemic	Volume depletion	Hypovolemia Hemorrhage	↓ Volume (↓ CO)	• Fluid resuscitation • Pressors
Obstructive	Impairment of cardiopulmonary blood flow	Pulmonary Embolism Cardiac Tamponade Tension Pneumothorax	↓ CO	• Fluid resuscitation • Relieving the obstruction

Fluid Resuscitation: Blood

- There are three types of fluids that can be used:
- Blood Products:
 - Packed RBCs: consider using for mass transfusion protocol in hemorrhagic shock
 - Fresh Frozen Plasma: consider again for mass transfusion protocol
 - Can also use to combat DIC, which can occur with septic shock
 - Platelets: treat bleeding
 - Cryoprecipitate: for fibrinogen repletion



- (2022) Sepsis and Septic Shock. Harrison's Principles of Internal Medicine.
- UpToDate: Use of blood products in the critically ill

Fluid Resuscitation: IV Fluids

- **Crystalloids: have small molecules**
 - 0.9% Normal Saline:
 - Concern of potential metabolic acidosis from hyperchloremia in large fluid resuscitation
 - Ringer's Lactate Solution:
 - LR has a lower chloride equivalent
 - No clear cut better crystalloid when comparing LR to NS
 - Balanced salt solution (PlasmaLyte): matches the electrolyte content of plasma but expensive
- **Colloids: have larger molecules that cannot escape the plasma (proteins, dextran)**
 - Albumin
- Generally, no differences have been found in patient outcomes between crystalloid and colloid solutions

Fluid Resuscitation: IV Fluids

Table 1. Types and Compositions of Resuscitation Fluids.*													
Variable	Human Plasma	Colloids									Crystalloids		
		4% Albumin	Hydroxyethyl Starch						4% Succinylated Modified Fluid Gelatin	3.5% Urea-Linked Gelatin	0.9% Saline	Compounded Sodium Lactate	Balanced Salt Solution
			10% (200/0.5)	6% (450/0.7)	6% (130/0.4)		6% (130/0.42)						
Trade name		Albumex	Hemohe	Hextend	Voluven	Volulyte	Venofundin	Tetraspan	Gelofusine	Haemaccel	Normal saline	Hartmann's or Ringer's lactate	PlasmaLyte
Colloid source		Human donor	Potato starch	Maize starch	Maize starch	Maize starch	Potato starch	Potato starch	Bovine gelatin	Bovine gelatin			
Osmolarity (mOsm/liter)	291	250	308	304	308	286	308	296	274	301	308	280.6	294
Sodium (mmol/liter)	135–145	148	154	143	154	137	154	140	154	145	154	131	140
Potassium (mmol/liter)	4.5–5.0			3.0		4.0		4.0		5.1		5.4	5.0
Calcium (mmol/liter)	2.2–2.6			5.0				2.5		6.25		2.0	
Magnesium (mmol/liter)	0.8–1.0			0.9		1.5		1.0					3.0
Chloride (mmol/liter)	94–111	128	154	124	154	110	154	118	120	145	154	111	98
Acetate (mmol/liter)						34		24					27
Lactate (mmol/liter)	1–2			28								29	
Malate (mmol/liter)								5					
Gluconate (mmol/liter)													23
Bicarbonate (mmol/liter)	23–27												
Octanoate (mmol/liter)		6.4											

* To convert the values for potassium to milligrams per deciliter, divide by 0.2558. To convert the values for calcium to milligrams per deciliter, divide by 0.250. To convert the values for magnesium to milligrams per deciliter, divide by 0.4114.

Fluid Resuscitation

- Which Fluids:
 - Consider blood transfusion with a Hgb of < 7 g/dL
 - Colloids: albumin does not have any long-term benefits over crystalloids
 - Crystalloids: No clear cut better crystalloid when comparing LR to NS
 - Normal Saline: run the risk of hyperchloremic metabolic acidosis due to a higher Cl^- quantity
 - Ringer's Lactate: closer physiologic concentrations, but can run a risk of hyponatremia
- How Much?
 - Surviving Sepsis Campaign: **>30 mL/kg of crystalloids within the first 3 hours**
 - Example: a 70 kg individual should receive $\sim 2,100$ mL (2 L) of fluids
- How to assess?
 - Clinical: HR, BP, Urine Output, Skin Turgor, Mucous Membranes
 - Vitals: MAP >65 mmHg; SVO₂ $>70\%$

Medication: Vasoactive Agents

- While addressing the underlying cause for shock is the best treatment modality, vasoactive agents can help restore blood pressure
- Norepinephrine is a first line agent for septic shock
 - Dobutamine or another inotrope should be used to cardiogenic shock
- Complications:
 - Extravasation: vasoconstriction of nearby connective tissue can cause skin necrosis
 - Hypoperfusion: prolonged vasoconstriction can produce inadequate perfusion of the extremities, mesenteric organs, and kidneys

Medication: Vasoactive Agents

DRUG	Clinical Indication	Receptor Binding	Major Effect	Notes and AE
Norepinephrine	Shock (distributive, cardiogenic, mixed)	$\alpha_1 \gg \beta_1 > \beta_2$	$\uparrow$ SVR $\uparrow$ CO	Usually first-choice pressor AE: Tachyarrhythmias, peripheral (digital) ischemia
Phenylephrine	Shock (distributive, hypovolemic)	α_1	$\uparrow$ SVR	Useful in tachyarrhythmias AE: Reflex bradycardia, severe peripheral and visceral vasoconstriction
Vasopressin	Shock (distributive, cardiogenic)	V_1, V_2	$\uparrow$ SVR	Add to norepinephrine in septic shock AE: Arrhythmias, cardiac ischemia, peripheral and splanchnic vasoconstriction
Epinephrine	Shock (anaphylactic, cardiogenic, distributive), cardiac arrest, bronchospasm	$\alpha_1 > \beta_1 > \beta_2$	$\uparrow$ HR $\uparrow$ SVR $\uparrow$ CO	First line for anaphylaxis and cardiac arrest AE: Ventricular arrhythmias, cardiac ischemia
Dopamine	Bradycardia Shock (cardiogenic, distributive)	$D_1 \gg \beta_1 > \alpha_1 > \beta_2$	$\uparrow$ CO mild $\uparrow$ SVR	Not first-line treatment AE: Ventricular arrhythmias, cardiac ischemia, tissue ischemia/gangrene
Dobutamine	Cardiogenic shock	$\beta_1 \gg \beta_2 > \alpha_1$	$\uparrow$ CO	Inotrope, not vasopressor Can cause hypotension AE: Ventricular arrhythmias, cardiac ischemia

Shock: Summary

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Shock Inadequate organ perfusion and delivery of nutrients necessary for normal tissue and cellular function. Initially may be reversible but life threatening if not treated promptly.

TYPE	CAUSED BY	MECHANISM	SKIN	CVP	PCWP	CO	SVR	SVO ₂
Hypovolemic shock	Hemorrhage, dehydration, burns	Volume depletion	Cold, clammy	↓	↓	↓	↑	↓
Cardiogenic shock	MI, HF, valvular dysfunction, arrhythmia	Left heart dysfunction		↑	↑	↓	↑	↓
Obstructive shock	PE, tension pneumothorax	Impeded cardiopulmonary blood flow		↑	↓	↓	↑	↓
	Cardiac tamponade			↑	↑	↓	↑	↓
Distributive shock	Sepsis (early), anaphylaxis	Systemic vasodilation	Warm, dry	↓	↓	↑	↓	↑
	CNS injury			↓	↓	↓	↓	normal/↑

↓ = 1° disturbance driving the shock.

Shock Summary : Cont'd

- Cardiac Shock: cardiac pumping ability is compromised
 - Cardiac output decreases
 - Causes: arrhythmias, valve malfunction, coronary occlusions, myocardial infarction
- Hypovolemic Shock: significant hemorrhage or fluid depletion
 - Decreased circulating blood volume causing decreased cardiac output (decreased stroke volume)
 - Causes: hemorrhage; inadequate fluid intake, severe burns, chronic diarrhea, prolonged vomiting
- Obstructive Shock: reduced cardiac filling due to obstruction or restriction of the heart
 - Decreased stroke volume
 - Causes: cardiac tamponade (prevents adequate diastolic filling); pulmonary embolus (limits blood entering the LA)

Shock Summary : Cont'd

- **Distributive Shock:** decrease in systemic vascular resistance
 - Anaphylactic Shock: due to severe allergic reaction
 - Mediated by histamine, prostaglandins, leukotrienes, and bradykinin causing vasodilation & increased microvascular permeability
 - Septic Shock: due to infection (typically Gram negative)
 - Mediated by endotoxins releasing a nitric oxide synthase causing vasodilation
 - Neurogenic Shock: due to inability of CNS to deliver sympathetic vasoconstriction
 - Occurs with deep anesthesia or a reflex response to traumatic injuries

Question 1:

An 18 year old female presents to your office severely fatigued. She reports that she has had episodes of diarrhea and vomiting for the past 4 days. Vitals reveal: temperature 102.7°F, BP 75/50mm Hg, and HR 110/min. Hemodynamic measurement reveals a ↑SVR. Which of the following is the most likely diagnosis:

- A. Cardiogenic Shock
- B. Obstructive Shock
- C. Hypovolemic Shock
- D. Distributive Shock

Question 1: Answer

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- A. Cardiogenic Shock
- B. Obstructive Shock
- C. Hypovolemic Shock**
- D. Distributive Shock

Rationale:

- Diarrhea and vomiting, the pt is unable to retain or replenish her fluids.
- Elevated temperature: typically can be seen in septic shock and hypovolemic (due to illness)
- Since SVR is elevated, this cannot be septic shock

Question 2:

An 85 year old man presents to the ED with fatigue. He has a past medical history of heart failure, HTN, and gout. History reveals nonadherence with his medications due to difficulty picking up his prescriptions. Vitals reveal: temperature 98.7°F, BP 74/41 mm Hg, HR 127/min, and RR 23/min. His jugular veins are distended. Which of the following hemodynamic measurements do you expect to find on further examination?

- A. ↓CO ↑SVR ↑PCWP
- B. ↓CO ↓SVR ↓PCWP
- C. ↓CO ↑SVR ↓PCWP
- D. ↑CO ↑SVR ↓PCWP
- E. ↑CO ↑SVR ↑PCWP

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- C. ↓CO ↑SVR ↓PCWP
- D. ↑CO ↑SVR ↓PCWP
- E. ↑CO ↑SVR ↑PCWP

Rationale:

- This patient is in cardiogenic shock! We expect to see decreased cardiac output, and reflexive vasoconstriction
- PCWP will be increased, as there is a backup of fluid in the LA due to LV failure
- PMHx: patient has known history of heart failure with a nonadherence to medication
- While obstructive shock (PE) may appear similar, PMHx points to heart failure

Question 3:

A 56-year-old man is brought to the emergency department 30 minutes after slicing his arm while chopping wood, leading to significant blood loss. His blood pressure is 92/58 mm Hg. On examination, he appears lethargic and pale. His extremities are cool to the touch.

Choice	Heart rate	Total peripheral resistance	Renin	Cardiac contractility
A	↓	↓	↓	↓
B	↓	↑	↑	↑
C	↑	↓	↑	↑
D	↑	↑	↑	↓
E	↑	↑	↑	↑

Which set of findings would most likely represent the body’s physiologic reaction within the first 30 minutes of this patient’s injury?

- A. A
- B. B
- C. C
- D. D
- E. E

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- A. A
- B. B
- C. C
- D. D
- E. E

Rationale:

- This patient is in hypovolemic shock
- We would expect to see a drop in BP, with sympathetic reflexive activation (↑ SVR, ↑ HR)
- This causes increase in **heart rate and contractility; NOT CO**
- Renin will be released in an attempt to raise BP (RAAS System)

Question 4:

A 75-year-old nursing home resident is brought to the emergency department because of confusion. He developed a productive cough with green sputum several days ago. Temperature is 38.9°C (102°F), pulse is 110/min, and blood pressure is 90/60 mm Hg. Results of laboratory studies show a WBC count of 14,000/mm³, a platelet count of 50,000/mm³, and elevated serum lactic acid.

Choice	CVP	PCWP	CO	SVR
A	↓	↓	↓	↑
B	↓	↓	↑	↓
C	↓	↓	↓	↓
D	↑	↑	↑	↓
E	↑	↑	↓	↑

Which set of physiologic parameters is most likely for this patient?

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C	↓	↓	↓	↓
D	↑	↑	↑	↓
E	↑	↑	↓	↑

Which set of physiologic parameters is most likely for this patient?

- A. A
- B. B
- C. C
- D. D
- E. E

Rationale:

- This patient is in septic shock
- Primary disturbance is a **DECREASE** in SVR
- This causes increase in heart rate, since we are not losing fluid, we will see an increase in CO
- PCWP and CVP will not increase, the heart is functioning fine

Question 5:

You are an intern on call and are paged to see a 39 year old male. He has a past medical history of cystic fibrosis and had a bilateral lung transplant 2 years ago for which he is taking tacrolimus, azathioprine, and corticosteroids. VS reveal: temp 102°F, BP 70/35 mm Hg, HR 134/min, and respirations 18/min.

What is the primary disturbance causing this patient's shock?

- A. Decreased cardiac output due to decreased cardiac contractility
- B. Decreased cardiac output due to decreased diastolic filling
- C. Loss of fluids due to poor fluid resuscitation
- D. Decrease in systemic vascular resistance caused by reaction to lipopolysaccharide by toll-like receptors
- E. Decrease in systemic vascular resistance due to histamine release caused by autoimmune hypersensitivity
- F. Decrease in systemic vascular resistance due to unopposed parasympathetic tone

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Thank You!

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